Discrete Structure (CST-002) - Unit 2

Unit 2 - Propositional Calculus & Lattices - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: propositional logic (truth tables, equivalences, normal forms) and lattices - posets where every pair has a meet and a join.

Key ideas: A **proposition** is a statement that is true or false. Learn the connectives: AND, OR, NOT, implication (p → q is false only when p is true and q is false), biconditional. A formula true in every row is a **tautology**, false in every row a **contradiction**. Two formulas are **logically equivalent** when their truth tables match - memorise De Morgan's laws, distributive laws, and p → q ≡ ¬p ∨ q. **DNF** = OR of ANDs, **CNF** = AND of ORs. A **lattice** is a poset where every pair has a least upper bound (join) and greatest lower bound (meet). Then: **bounded** (has 0 and 1), **distributive**, **complemented** (every element has a complement) - all three together = **Boolean algebra**.

Exam pointers: truth-table questions are free marks - be systematic. For lattices, practice checking "is this poset a lattice?" by finding joins/meets of every pair from the Hasse diagram, and identifying sublattices and special lattices.

Extra note - Special lattices - 1-minute notes:

Lattice = a poset in which every pair of elements has a **meet (greatest common lower bound)** and a **join (least common upper bound)**.

- **Bounded lattice:** has a greatest element (1) and a least element (0). Example: in the power set P(S), S is the greatest and the empty set is the least.
- **Distributive lattice:** a AND (b OR c) = (a AND b) OR (a AND c) - the operation distributes both ways. The power set with union/intersection is distributive.
- **Complemented lattice:** every element has a complement (x OR x' = 1, x AND x' = 0). In a power set, the complement is the set difference.

Exam tip: from the Hasse diagram, check every pair for a meet and a join. If even one pair has no meet or no join, it is NOT a lattice.

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